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Samsung Global Sales Data Analysis Project using Excel

About the dataset:

The Samsung Global Sales dataset is from Kaggle. It is a synthetic dataset representing global product sales transactions for Samsung across 52 countries from 2021 to 2024. This dataset will serve as practice and a project for my portfolio.

Link: https://www.kaggle.com/datasets/ashyou09/samsung-global-product-sales-dataset?resource=download

The tool I used for this project is Microsoft Excel. Unlike my previous project where I used SQL for data cleaning and EDA, this dataset was already clean and complete, so I went straight to the analysis using Excel PivotTables. I then built a dashboard to visualize the insights I found.

sale_id	sale_date	year	quarter	month	country	region	city	product_name	category	storage	color	is_5g	unit_price_usd	discount_pct
SAMS-00000001	1/1/2021	2021	Q1	January	Argentina	South America	Buenos Aires	Samsung Galaxy Tab S9 Ultra	Galaxy Tab	1 TB	Graphite	No	1246.77	0
SAMS-00000002	3/23/2021	2021	Q1	March	Argentina	South America	Buenos Aires	Samsung Galaxy S23	Galaxy S	1 TB	Titanium Violet	Yes	809.84	10
SAMS-00000003	5/22/2021	2021	Q2	May	Argentina	South America	Buenos Aires	Samsung Galaxy A34 5G	Galaxy A	128 GB	Awesome Black	Yes	410.19	12
SAMS-00000004	7/26/2021	2021	Q3	July	Argentina	South America	Buenos Aires	Samsung T55 27-inch FHD	Monitor	N/A	Black	No	242.69	0
SAMS-00000005	9/2/2021	2021	Q3	September	Argentina	South America	Buenos Aires	Samsung Galaxy Z Fold 4	Galaxy Z	256 GB	Cream	Yes	1562.98	0

Figure 1. Snapshot of the Dataset

Since the dataset was already clean and complete upon download, no heavy data cleaning was needed. All columns were properly filled with no, duplicates, or formatting errors. Empty cells from customer_rating was changed into nulls. The dataset contains the following fields: sale_id, sale_date, year, quarter, month, country, region, city, product_name, category, storage, color, is_5g, unit_price_usd, discount_pct, units_sold, discounted_price_usd, revenue_usd, currency, fx_rate_to_usd, revenue_local_currency, sales_channel, payment_method, customer_segment, customer_age_group, previous_device_os, customer_rating, and return_status. I proceeded directly to the EDA stage without any cleaning steps.

units_sold	discounted_price_usd	revenue_usd	currency	fx_rate_to_usd	revenue_local_currency	sales_channel	payment_method	customer_segment	customer_age_group	previous_device_os	customer_rating	return_status
1	1246.77	1246.77	ARS	907	1130820.39	E-commerce Platform	Samsung Pay	Business	45-54	N/A	3.2	Kept
2	728.86	1457.72	ARS	907	1322152.04	Authorized Reseller	Net Banking	Government	55+	Feature Phone	null	Kept
6	360.97	2165.82	ARS	907	1964398.74	Corporate / B2B	Gift Card	Individual	25–34	New User	3.5	Kept
3	242.69	728.07	ARS	907	660359.49	Third-Party Retailer	BNPL (Buy Now Pay Later)	Enterprise	55+	N/A	4	Kept
2	1562.98	3125.96	ARS	907	2835245.72	Authorized Reseller	Gift Card	Business	55+	Android (Other)	3	Kept
2	204.71	409.42	ARS	907	371343.94	E-commerce Platform	Net Banking	Government	25–34	N/A	2.6	Kept
4	103.66	414.64	ARS	907	376078.48	Carrier Store	Net Banking	Enterprise	55+	N/A	4.5	Kept
2	173.77	347.54	ARS	907	315218.78	Third-Party Retailer	Debit Card	Education	25–34	Feature Phone	null	Kept
1	157.9	157.9	ARS	907	143215.3	Online (Samsung.com)	EMI / Installment	Individual	18–24	N/A	null	Kept

Figure 2. Cleaned Dataset Snapshot

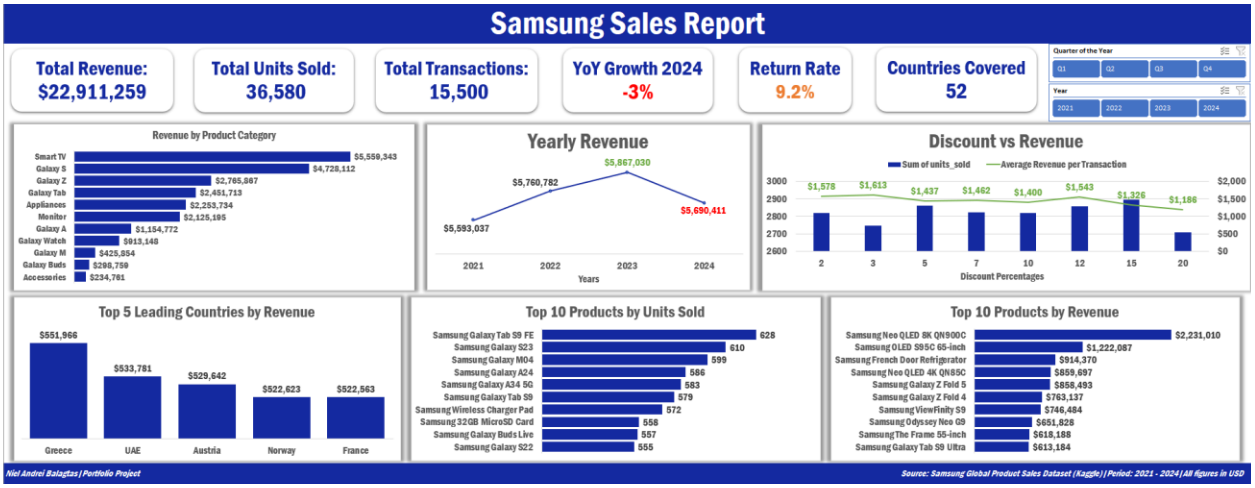

Figure 3. Samsung Sales Dashboard

For the EDA, I tried to answer common business questions using Excel PivotTables and charts. Below are the questions I explored and the insights I found from the data.

1. Top 5 leading countries per month by revenue
2. Highest revenue item
3. Highest sold item
4. Top returned item
5. Which segment spends more
6. Which age group spends more
7. Top 10 countries that sell 5G items
8. Which sales channel produces the highest revenue
9. Customer rating vs. return status
10. Discount vs. revenue
11. Which product category sells more
12. Yearly revenue comparison
13. Most returned item

The dataset covers the years 2021 to 2024. The total revenue is **$22,911,259**, with **36,580 total units sold** and **15,500 total transactions** across **52 countries**. The year-over-year growth for 2024 was **-3%**, meaning revenue slightly dropped compared to the peak year of 2023. The overall return rate is **9.2%**, which is something worth watching to better understand customer satisfaction.

Looking at revenue by product category, **Smart TV leads with $5,559,343**, followed by the **Galaxy S series at $4,728,112**. Accessories recorded the lowest revenue at $234,781. Even though Smart TV tops the chart, this is largely driven by its high unit price. The Galaxy S series performing strongly shows consistent demand for Samsung flagship smartphones. Cookie being the lowest in revenue in the cafe project is similar here — budget accessories bring in the least money even if they sell in volume.

For yearly revenue, **2023 was the best-performing year at $5,867,030**. Revenue grew steadily from 2021 ($5,593,037) to 2022 ($5,760,782) and peaked in 2023, before dipping to **$5,690,411 in 2024**. The steady growth from 2021 to 2023 is a good sign for the business, though the 2024 decline is something that should be looked into.

The **top 5 countries by revenue** are Greece ($551,966), UAE ($533,781), Austria ($529,642), Norway ($522,623), and France ($522,563). These countries consistently lead in total revenue and can be targeted for stronger marketing campaigns. For products, the **Samsung Galaxy Tab S9 FE leads in units sold with 628 units**, while the **Samsung Neo QLED 8K QN900C leads in revenue at $2,231,010**. This shows that premium products earn more money per transaction even if they are not the most sold item.

The discount vs. revenue analysis shows that a **2% discount produces the highest average revenue per transaction at $1,578**, while a **20% discount drops that figure to $1,186**. This means that giving bigger discounts does not lead to higher revenue per transaction. Heavy discounting may actually reduce overall revenue, so discount strategies should be used carefully.

Note: I have done the EDA in Excel using PivotTables and the Dashboard in Excel (check file).

To conclude this Data Analysis project, I can say that the Samsung global sales business is generally on a healthy track based on the data. There is steady revenue growth from 2021 to 2023, with premium products like Smart TVs and Galaxy S flagships driving the most revenue. However, the -3% YoY decline in 2024 and the 9.2% return rate are areas that need attention. Heavy discounting appears to lower average revenue per transaction, suggesting the business should be cautious about deep discounts. The top countries by revenue such as Greece, UAE, and Austria present strong opportunities for targeted growth strategies.

I just want to note that this is a synthetic dataset and this is how I would analyze real data as well. I made use of Microsoft Excel including PivotTables, Excel charts, and slicers to complete this project.